TM Forum Component

Resource Catalog Management

TMFC010

Maturity Level: General Availability (GA)	**Team Approved Date: 11-May-2026**
Release Status: Production	**Approval Status: TM Forum Approved**
Version 1.4.2	**IPR Mode: RAND**

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TMFC010 – Resource Catalog Management

1. Overview

Component Name	ID	Description	ODA Function Block
Resource Catalog Management	TMFC010	The Resource Catalog Management component is responsible for organizing the collection of resource specifications that identify and define all requirements for a resource. The Resource Catalog Management component has the functionality that enables presentation of a customer-facing view, so users are able to browse and select resources they need, as well as a technical view to enable definition and setup of resources contained in the resource catalog. Additional functionalities include capturing new resource specifications, managing resources (registering assets and components and identifying and mapping connections/relationships), administering the lifecycle of resources, describing relationships between resources, reporting on resources and changes to their attributes, and facilitating easy access to identify and assign resources.	Production

2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM L2 - SID ABEs links

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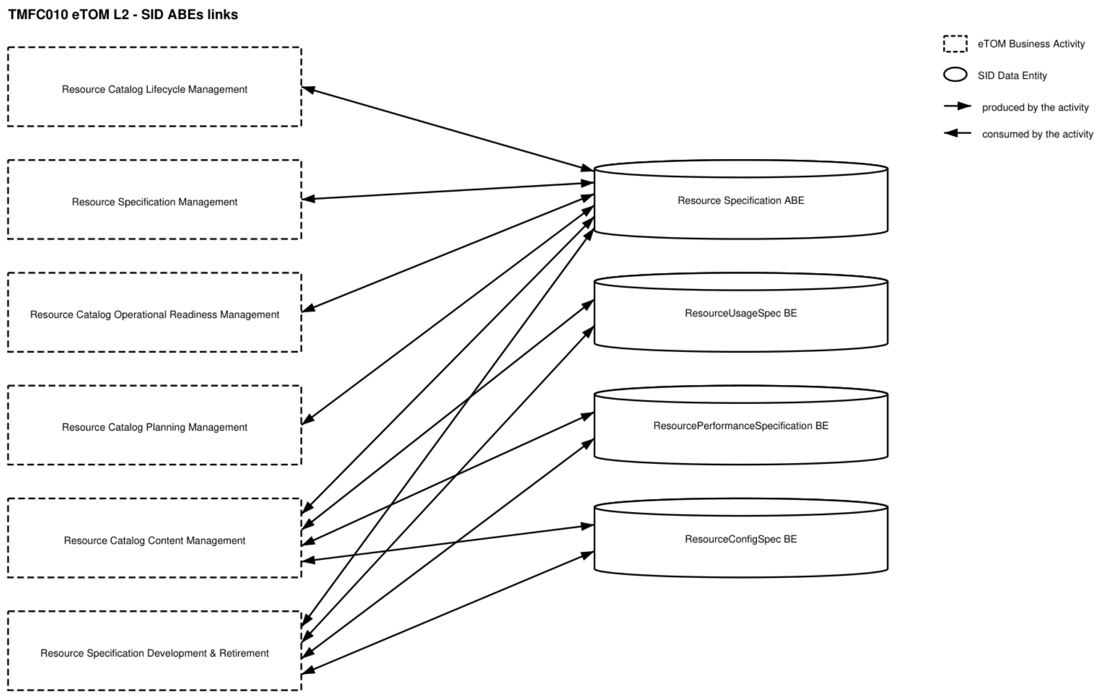

2.2. eTOM business activities

eTOM business activities this ODA Component is responsible for:

Identifier	Level	Business Activity Name	Description
1.5.15	L2	Resource Catalog Lifecycle Management	Catalog Lifecycle Management business process covers a set of business activities that enable manage the lifecycle of an organizations catalog from design to build according to defined requirements. / Catalog Lifecycle Management proves the overarching governance to manage all the stages in the realization and operationalization of the Product/Service/Resource Catalog in support of the organizations business goals.

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1.5.16	L2	Resource Catalog Operational Readiness Management	Resource Catalog Operational Readiness Management business process establishes and administers the support needed to operationalize Resource catalogs for ongoing day-to-day business needs. / These business activities implement the Resource Catalog through Release and Deploy business activities.
1.5.17	L2	Resource Catalog Content Management	Resource Catalog Content Management business process define and provide the business activities that support the day-to-day operations of Resource Catalogs in order to realize the business operations goals. / Resource Catalog Content Management business processes include administering the Resource Catalog instance in production, maintaining catalog entries, assuring catalogs, managing catalog access, managing entry lifecycle through versioning, handling catalog entity entry and changes, supporting distribution of catalogs as needed, and supporting user-facing activities.
1.5.18	L2	Resource Catalog Planning Management	Resource Catalog Planning Management business process covers a set of business activities that understand and enable establish the plan to define, design and operationalize a catalog in order to meet the needs and objectives of Resource cataloging. / The Resource Catalog Planning Management business process ensure that the organization is able to identify the most appropriate scheme and goal for it catalog. It includes designing the Catalog plan and developing the specification according to Resource management requirement.

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1.5.19	L2	Resource Specification Management	Resource Specification Management business process leverages captured resource requirements to develop, master, analyze, and update documented standard conditions that must be satisfied by a resource design and/or delivery. / Resource Specifications Management can result in establishing in a centralized way, technical (know-how) standards. Such standards provide the organization with a means to control and approve the values and inputs of specification through structure, review, approval and distribution processes to stakeholders and suppliers.
1.5.3	L2	Resource Specification Development & Retirement	Resource Specification Development & Retirement processes develop new, or enhance existing technologies and associated resource types, so that new Products are available to be sold to customers. They use the capability definition or requirements defined by Resource Strategy & Planning They also decide whether to acquire resources from outside, taking into account the overall business policy in that respect. These processes also retire or remove technology and associated resource types, which are no longer required by the enterprise. / Resource types may be built, or in some cases leased from other parties. To ensure the most efficient and effective solution can be used, negotiations on network level agreements with other parties are paramount for both building and leasing. / These processes interact strongly with Product and Engaged Party Development processes.

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1.5.3.4	L3	Develop Detailed Resource Specifications	The Develop Detailed Resource Specifications processes develop and document the detailed resource-related technical, performance and operational specifications, and manuals. These processes develop and document the required resource features, the specific technology requirements and selections, the specific operational, performance and quality requirements and support activities, any resource specific data required for the systems and network infrastructure. The Develop Detailed Resource Specifications processes provide input to these specifications. The processes ensure that all detailed specifications are produced and appropriately documented. Additionally, the processes ensure that the documentation is captured in an appropriate enterprise repository.

2.3. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE L1	SID ABE L1 Definition	SID ABE L2 (or set of BEs)	SID ABE L2 Definition

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Resource Specification	The Resource Specification ABE contains entities that define the shared characteristics and behavior of each of the three types of Resource entities. This enables multiple instances to be derived from a single specification entity. In this derivation, each instance will use the shared characteristics and behavior defined in its associated template. A ResourceSpecification might be Physical such as a handset, Logical such as an IP address or Compound i.e. composed of Logical and / or Physical Resources such as a Box with its firmware.		*(no description available)*
Resource Performance	The Resource Performance ABE collects information used to correlate, consolidate, and validate various performance statistics and other operational characteristics of Resources. Each of these entities also enables network performance assessment against planned goals, performs various aspects of trend analysis, including error rate and cause analysis and Resource degradation. Entities in this ABE also includes statistics defining Resource loading, and traffic trend analysis.	ResourcePerformanceSpecification	A measure of the manner in which a Resource is functioning.

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Resource Usage	The Resource Usage ABE represents the specification of Resource Usage types and collects Resource Usage records with their characteristics (a.k.a. consumption data). The entities in this ABE provide physical, logical, and network usage information.	ResourceUsageSpec	This Entity defines the attributes that are common to all resources that make up a Product or used to configure a Service or to support an enterprise infrastructure.
Resource Configuration	The Resource Configuration ABE represents the specification of the different possible Resource configurations as well as the Resource Configurations instantiated for a Resource.	ResourceConfigSpec	The definition of how a Resource operates or functions in terms of CharacteristicSpecification(s) and related ResourceSpec(s).

2.4. Functional Framework Functions

Function ID	Function Name	Aggregate Function Level 1	Aggregate Function Level 2	Function Description
737	Resource Capability Specification Management	Resource Capability Management	Resource Specification Capability Development	This function involves the creation, editing, storage and retrieval of capability specifications. The capability specifications represent the general, common and invariant characteristics of resource that may be realized in more than one type of specific resource. Examples of capability are Layer2, Data, radio and Transport.

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467	Resource Data Transformation/Parsing Rules Configuration	Resource Specification Management	Resource Specification Development	Resource Data Transformation/Parsing Rules Configuration provides tools to set up and maintain resource data parsing rules.
951	Resource Catalog Entities Management	Resource Specification Management	Resource Specification Development	Resource Catalog Entities Management identifies resource entities in a common Catalog Management from the Common Domain, or identifies a specific instance of a Catalog Management for resource entities.
996	Resource Task Item Policy Control Configuration	Resource Specification Management	Resource Specification Development	Defines and configures the policies which will be implemented during the Resource task item lifecycle.

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1083	Resource Specification Repository Management	Resource Specification Management	Resource Specification Development	Resource Specification Repository Management is able to create, modify and delete Resource Specification. / This includes the ability to manage the state of an entity during its lifecycle (e.g. planned, deployed, in operation, replaced by locked). / It includes Resource Specifications retrieval, integrity rules check and versioning management. / It also provides Product Specification and Offering views adapted to the different roles.
1088	Resource Specification to Supplier Product Specification Relationship Design	Resource Specification Management	Resource Specification Development	Resource Specification to Supplier Product Specification Relationship Design identifies, when it corresponds to equipment rented to a Supplier (devices, network equipments, hardware & software, etc).

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1089	Resource Specification Action Skill Design	Resource Specification Management	Resource Specification Development	Resource Specification Action Skill Design manages the links to the Skill catalog to identify for each type of resource, or of Action on a resource, which type of skill is necessary to make the intervention.
1082	Resource Specification Change Auditing	Resource Specification Management	Resource Specification Development	Resource Specification Change Auditing manages the implications of Resource Specifications changes to determine the consequences of any given change. Resource Specifications changes may impact other Resource Specifications and Resource Facing Service Specit supports / The function logs Resource Specifications changes and supports the analysis of relationships between Resource Specifications. / In addition, it tracks the history of changes in an easy and accessible manner.

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1064	Logical and Software Resources Designing	Resource Specification Management	Resource Specification Development	Logical and Software Resources Designing supports physical, logical, and software design of resources including definition of configuration variables and initial parameters.
1087	Resource Specification Design	Resource Specification Management	Resource Specification Development	Resource Specification Design manages Resource Specifications including constraints, characteristics and type of usages. / It specifies Physical, Logical and Compound Resource Specifications. / It includes resource types that as a Service Provider we don't own or commercialize (ex: mobile phones used by our customers).

3. TM Forum Open APIs & Events

The following part covers the APIs and Events; this part is split in 3: - List of Exposed APIs - This is the list of APIs available from this component. - List of Dependent APIs - In order to satisfy the provided API, the component could require the usage of this set of required APIs. - List of Events (generated & consumed) - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. API Context Diagram

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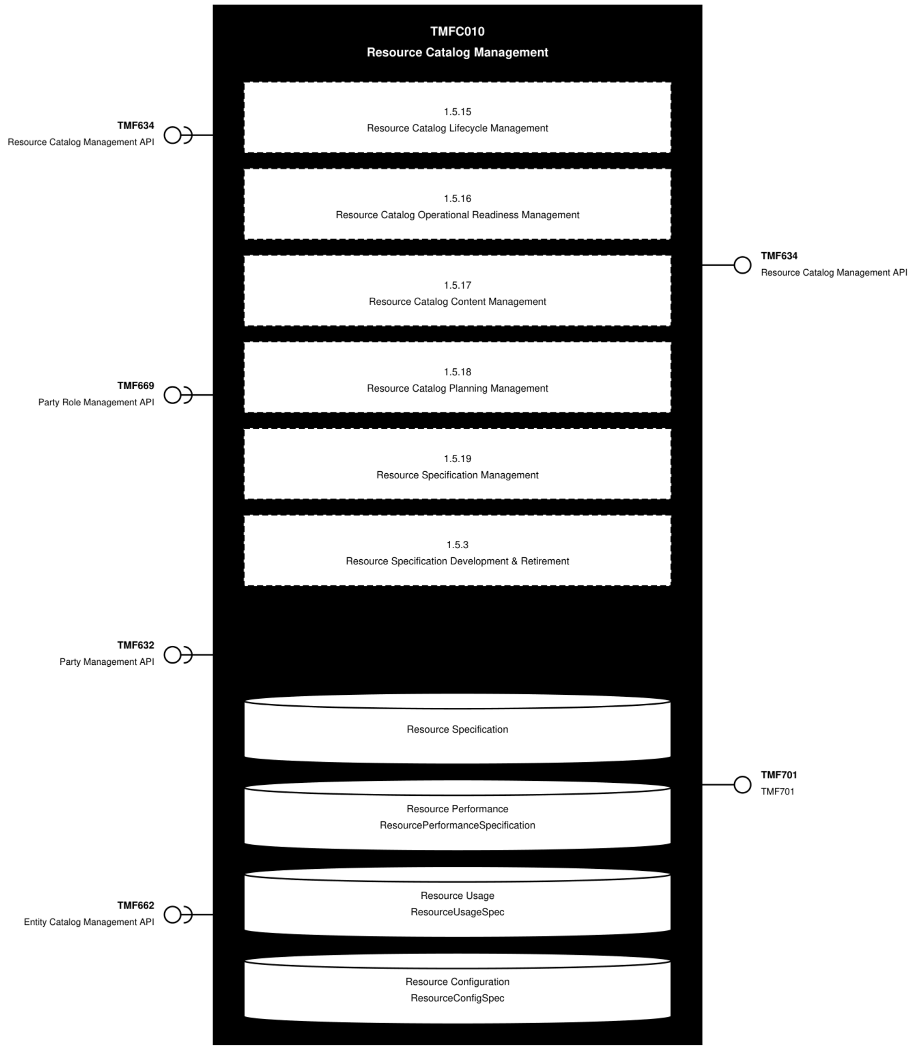

(SVG source: [TMFC010_API_Context.svg]() — hand-drawn rather than PlantUML for this one diagram, since PlantUML's automatic layout couldn't give each API its own straight, individually-anchored connector at a chosen height. Generated by the `component-specification-documentation` *skill's* `scripts/render_api_context_svg.py`*.)*

3.2. Exposed APIs

API ID	API Name	Mandatory / Optional	API Version	Resource	Operations
TMF634	Resource Catalog Management API	Mandatory	5	exportJob	GET, GET /id, POST, DELETE

TMF634	Resource Catalog Management API	Mandatory	5	importJob	GET, GET /id, POST, DELETE
TMF634	Resource Catalog Management API	Mandatory	5	resourceCandidate	GET, GET /id, POST, PATCH, DELETE
TMF634	Resource Catalog Management API	Mandatory	5	resourceCatalog	GET, GET /id, POST, PATCH, DELETE
TMF634	Resource Catalog Management API	Mandatory	5	resourceCategory	GET, GET /id, POST, PATCH, DELETE
TMF634	Resource Catalog Management API	Mandatory	5	resourceSpecification	GET, GET /id, POST, PATCH, DELETE
TMF634	Resource Catalog Management API	Mandatory	4	resourceCatalog	GET, GET /id, POST, PATCH, DELETE
TMF634	Resource Catalog Management API	Mandatory	4	resourceCategory	GET, GET /id, POST, PATCH, DELETE
TMF634	Resource Catalog Management API	Mandatory	4	resourceSpecification	GET, GET /id, POST, PATCH, DELETE
TMF634	Resource Catalog Management API	Mandatory	4	resourceCandidate	GET, GET /id, POST, PATCH, DELETE
TMF634	Resource Catalog Management API	Mandatory	4	exportJob	GET, GET /id, POST, DELETE
TMF634	Resource Catalog Management API	Mandatory	4	importJob	GET, GET /id, POST, DELETE
TMF701	TMF701	Optional	4	processFlow	POST, GET, GET /id, DELETE
TMF701	TMF701	Optional	4	taskFlow	PATCH, GET, GET /id

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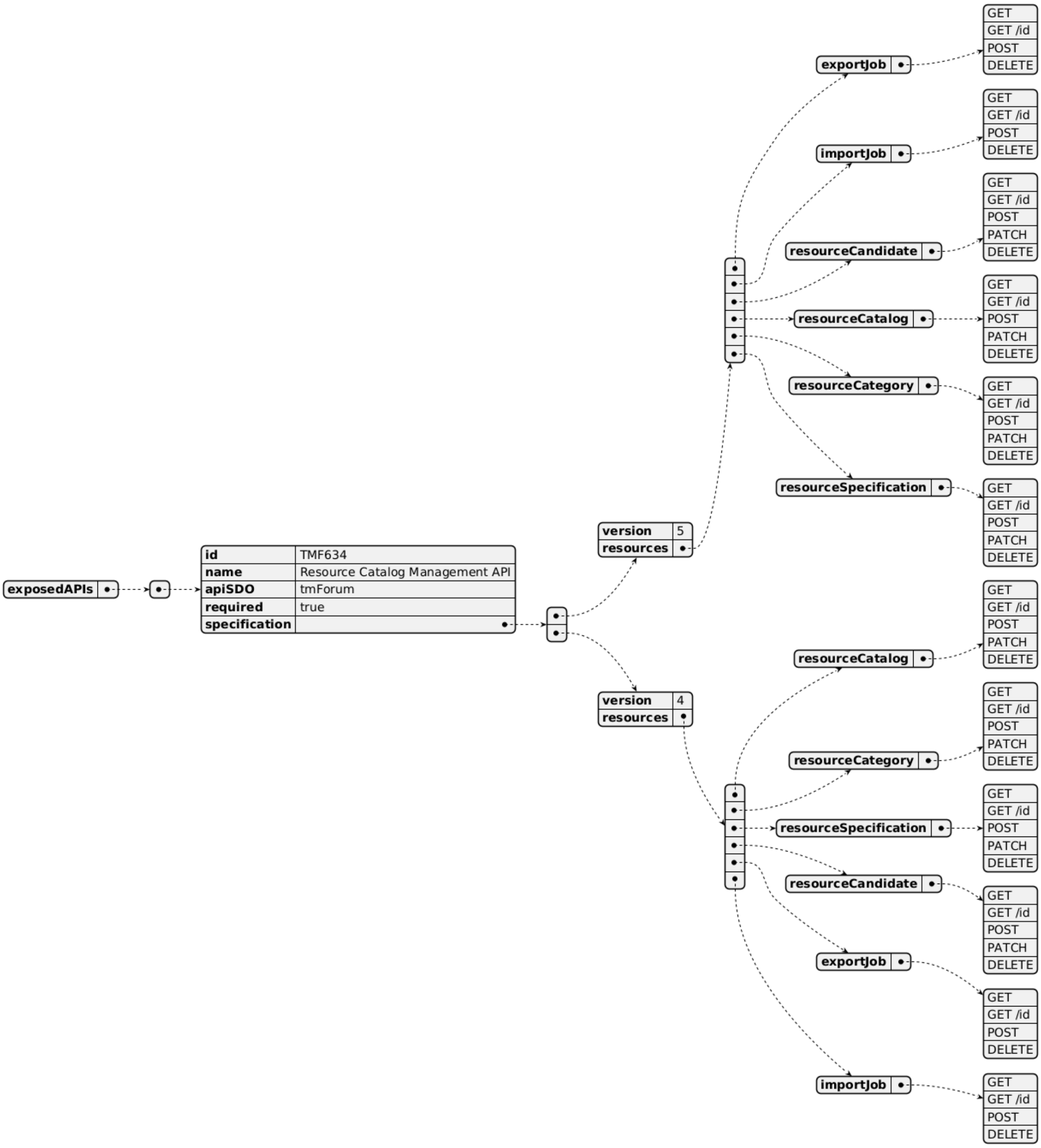

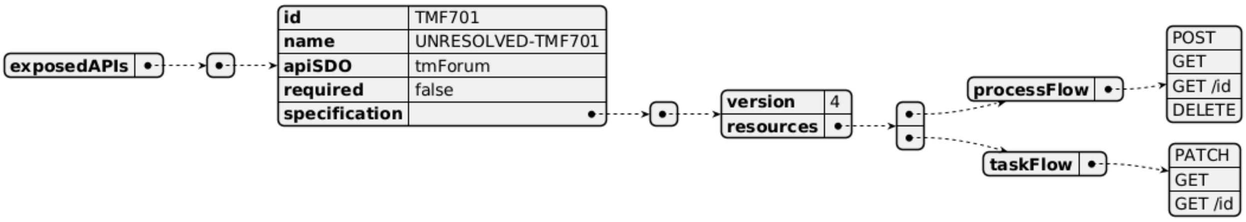

3.3. Dependent APIs

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API ID	API Name	API Version	Mandatory / Optional	Resource	Operations
TMF634	Resource Catalog Management API	5	Optional	exportJob	GET, POST, DELETE
TMF634	Resource Catalog Management API	5	Optional	importJob	GET, POST, DELETE
TMF634	Resource Catalog Management API	5	Optional	resourceCandidate	GET, GET /id, POST, PATCH, DELETE
TMF634	Resource Catalog Management API	5	Optional	resourceCatalog	GET, GET /id
TMF634	Resource Catalog Management API	5	Optional	resourceCategory	GET, GET /id, POST, PATCH, DELETE
TMF634	Resource Catalog Management API	5	Optional	resourceSpecification	GET, GET /id, POST, PATCH, DELETE
TMF634	Resource Catalog Management API	4	Optional	resourceCatalog	GET, GET /id
TMF634	Resource Catalog Management API	4	Optional	resourceCategory	GET, GET /id, POST, PATCH, DELETE
TMF634	Resource Catalog Management API	4	Optional	resourceSpecification	GET, GET /id, POST, PATCH, DELETE
TMF634	Resource Catalog Management API	4	Optional	resourceCandidate	GET, GET /id, POST, PATCH, DELETE
TMF634	Resource Catalog Management API	4	Optional	exportJob	GET, POST, DELETE
TMF634	Resource Catalog Management API	4	Optional	importJob	GET, POST, DELETE
TMF669	Party Role Management API	5	Optional	partyRole	GET, GET /id
TMF669	Party Role Management API	4	Optional	partyRole	GET, GET /id

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TMF632	Party Management API	5	Optional	individual	GET, GET /id
TMF632	Party Management API	5	Optional	organization	GET, GET /id
TMF632	Party Management API	4	Optional	individual	GET, GET /id
TMF632	Party Management API	4	Optional	organization	GET, GET /id
TMF662	Entity Catalog Management API	4	Optional	entitySpecification	GET, GET /id
TMF662	Entity Catalog Management API	4	Optional	associationSpecification	GET, GET /id

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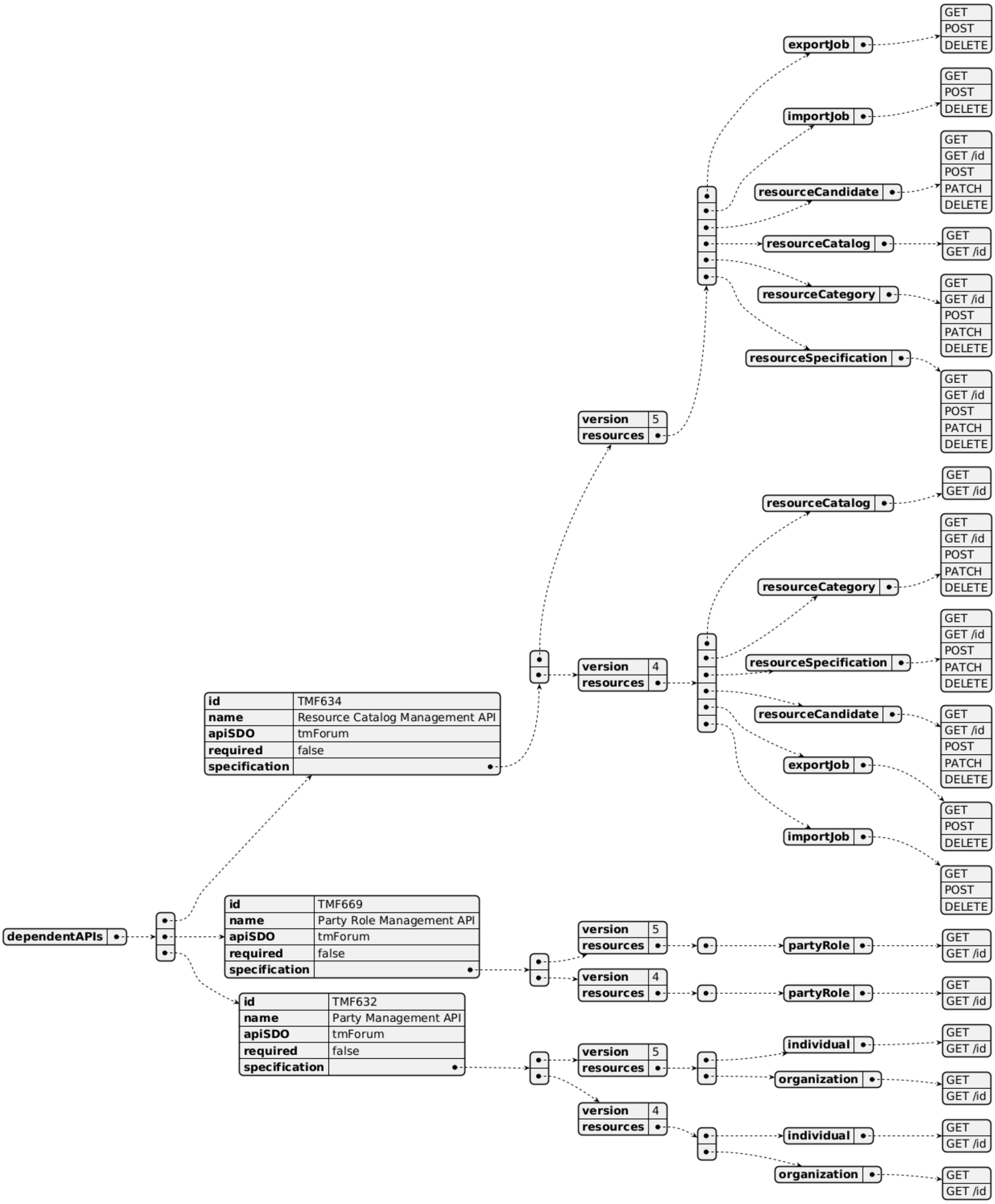
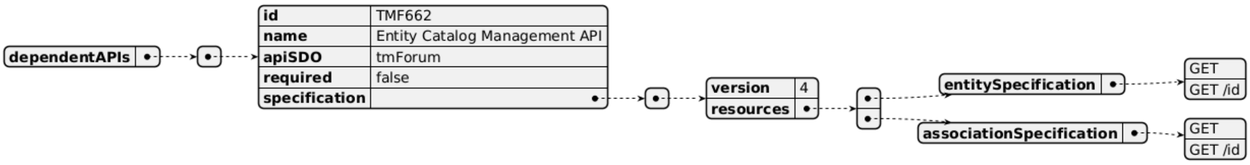

dependentAPIs
id TMF634
name Resource Catalog Management API
apiSDO tmForum
required false
specification
version 5
resources
exportJob
importJob
resourceCandidate
resourceCatalog
resourceCategory
resourceSpecification
version 4
resources
resourceCatalog
resourceCategory
resourceSpecification
resourceCandidate
exportJob
importJob
GET
GET /id
POST
PATCH
DELETE
id TMF669
name Party Role Management API
apiSDO tmForum
required false
specification
version 5
resources
partyRole
version 4
resources
partyRole
id TMF632
name Party Management API
apiSDO tmForum
required false
specification
version 5
resources
individual
organization
version 4
resources
individual
organization


dependentAPIs
id TMF662
name Entity Catalog Management API
apiSDO tmForum
required false
specification
version 4
resources
entitySpecification
associationSpecification
GET
GET /id

3.4. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.

Published Events

API ID	API Name	Event Resources
TMF634	Resource Catalog Management API	resourceCatalogCreateEvent resourceCatalogChangeEvent resourceCatalogDeleteEvent resourceCategoryCreateEvent resourceCategoryChangeEvent resourceCategoryDeleteEvent resourceCandidateCreateEvent resourceCandidateChangeEvent resourceCandidateDeleteEvent resourceSpecificationCreateEvent resourceSpecificationChangeEvent resourceSpecificationDeleteEvent importJobCreateEvent exportJobCreateEvent exportJobStateChangeEvent
TMF701	ProcessFlowManagement	processFlowCreateEvent processFlowStateChangeEvent processFlowStateDeleteEvent processFlowStateAttributeValueChangeEvent taskFlowCreateEvent taskFlowStateChangeEvent taskFlowDeleteEvent taskFlowAttributeValueChangeEvent taskFlowInformationRequiredEvent

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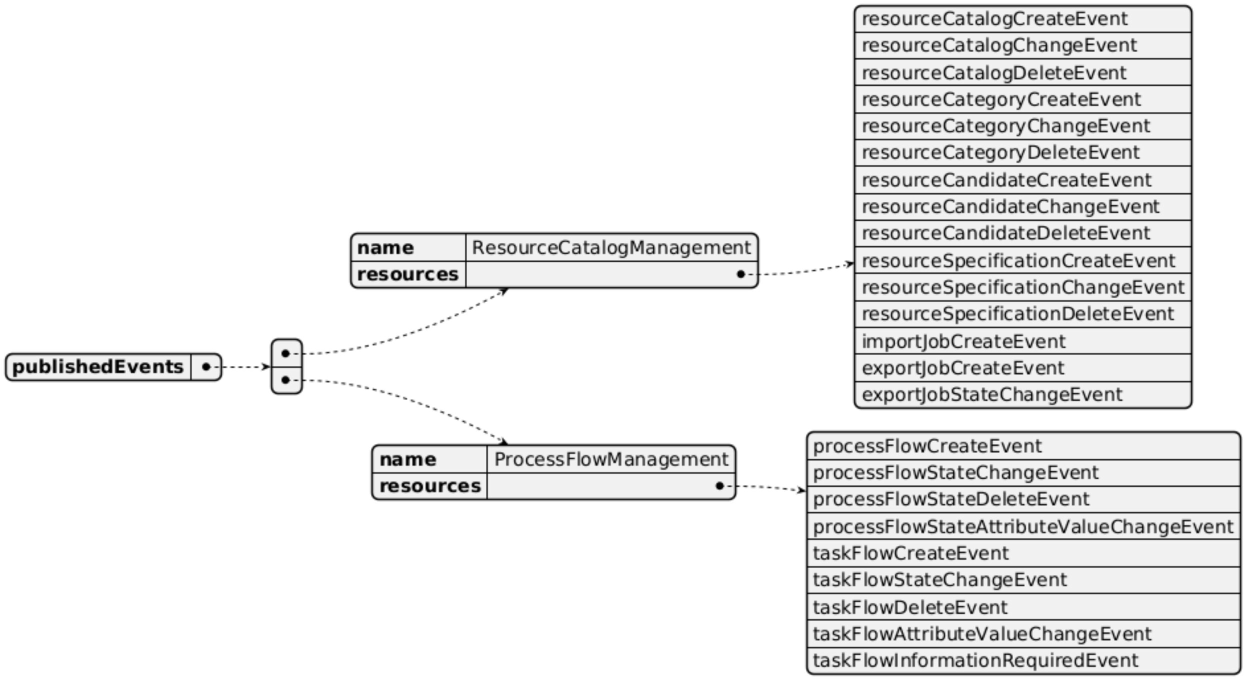
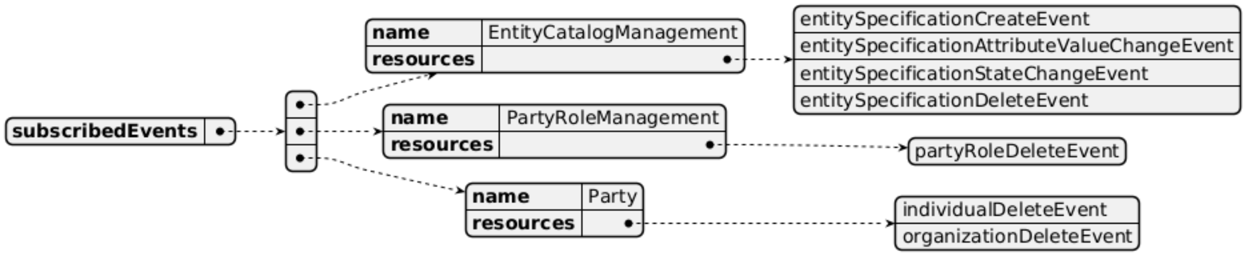


Subscribed Events

API ID	API Name	Event Resources
TMF662	Entity Catalog Management API	entitySpecificationCreateEvent entitySpecificationAttributeValueChangeEvent entitySpecificationStateChangeEvent entitySpecificationDeleteEvent
TMF669	Party Role Management API	partyRoleDeleteEvent
TMF632	Party Management API	individualDeleteEvent organizationDeleteEvent



4. Machine Readable Component Specification

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Refer to the ODA Component Directory on the TM Forum website for the machine-readable component specification files for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
eTOM	v23.0
SID	v25.0
Functional Framework	v23.0

5.2. Jira References

Open API - AP-4088 - Assess the addition of a ResourceUsage API with both ResourceUsage & ResourceUsageSpecification (DONE). The UsageSpecification is described in the Catalog Management components — at product, service, and resource levels. As part of the Resource Catalog Component specification, the question has arisen around the need to assess the addition of a ResourceUsage API with both ResourceUsage & ResourceUsageSpecification (feedback from Ludovic Robert and Kamal Maghsoudlou). This will be similar to the resource level like TMF727 (ServiceUsage) is for the service level. This is consistent with the SID ABE (resourceUsage & resourceUsageSpec).

Functional Framework - ISA-996 - Master Data Management has repurposed catalog related functions (BACKLOG). As part of reviewing the Product, Service and Resource Catalogs to align with the updates to the Functional Framework, we realized that a lot of functions have either been removed or been re-purposed by Master Data Management. Due to this, we had to make a decision to remove all these functions from the Components: removed deleted unclassified functions (1, 2, 5, 7, 9, 10, 12, 14, 15); removed functions that used to be related to Catalog management but now have been re-purposed by Master Data Management (3, 4, 6, 11, 13). Our concern is that we may be missing functions for the catalog due to re-use of Master Data Management.

5.3. Further resources

1. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

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6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date	Modified by	Description of changes
1.0.0	05-Aug-2022	Kamal Maghsoudlou, Gaetano Biancardi, Sylvie Demarest	Final edits prior to publication
1.1.0	06-Oct-2022	Elisabeth Andersson	Added support for federated catalogs and minor fixes.
1.1.1	27-Jul-2023	Ian Turkington	No content changed, simply a layout change to match template 3. Separated the YAML files to a managed repository.
1.1.1	14-Aug-2023	Amaia White	Final edits prior to publication
1.2.0	19-Apr-2024	Elisabeth Andersson	Update to latest template; aligned with 23.5 SID and eTOM (removed Resource Test specification BE and process, now managed by Resource Test Mgt); updated with Functional Frameworks 23.5 — added new functions under lvl2 Resource Specification Development (1064, 1082, 1083, 1087, 1088, 1089), updated the description of function 996, removed deleted unclassified functions (1, 2, 5, 7, 9, 10, 12, 14, 15), removed functions re-purposed by Master Data Management (3, 4, 6, 11, 13)
1.2.0	30-Apr-2024	Amaia White	Final edits prior to publication

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1.3.0	26-Nov-2024	Elisabeth Andersson	TMF688 removed from the core specification, moved to supporting functions; TMF672 removed from the core specification, moved to supporting functions; removed the minor versions of all APIs as per template
1.3.0	24-Dec-2024	Rosie Wilson	Final edits prior to publication
1.3.1	08-Jul-2025	Rosie Wilson	Updates to description per agreed YAML updated on 03 Jun 2025
1.3.1	15-Jul-2025	Rosie Wilson	Final edits prior to publication
1.3.2	13-Oct-2025	Gaetano Biancardi	Corrected naming format (without spaces) from "Resource Performance Specification BE" to "ResourcePerformanceSpecification BE"
1.3.2	19-Nov-2025	Rosie Wilson	Final edits prior to publication
1.4.0	05-Jun-2026	Hugo Vaughan (TM Forum)	Gen 5 API inclusion
1.4.0	25-Jun-2026	Sannah Sibaya	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by	Description of changes
Pre-production	05-Aug-2022	Goutham Babu	Initial release
Pre-production	07-Oct-2022	Alan Pope	Version 1.1.0
Pre-production	14-Aug-2023	Amaia White	Version 1.1.1
Production	06-Oct-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	30-Apr-2024	Amaia White	Version 1.2.0
Production	28-Jun-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status

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Pre-production	24-Dec-2024	Rosie Wilson	Version 1.3.0
Production	07-Mar-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	15-Jul-2025	Rosie Wilson	Version 1.3.1
Production	05-Sep-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	19-Nov-2025	Rosie Wilson	Version 1.3.2
Production	16-Jan-2026	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	25-Jun-2026	Sannah Sibaya	Version 1.4.0

6.2. Acknowledgements

This document was prepared by the members of the TM Forum Component and Canvas project team:

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